



TWOKEYS PROTOCOL

TwoKeys

Trustless deposits for marketplaces

One trust component for every marketplace and sector.

"You can disappear — but not with my money."

HackYeah 2026 · Superteam PL · Finance Without Intermediaries

BUYER DEPOSIT

D = 2,000

locked on reserve

SELLER STAKE

S = 2,000

locked first

PROGRAM-OWNED VAULT

P = 4,000

No one in the middle can touch it

No court, no escrow agent — just the rule

• LIVE ON SOLANA DEVNET

A deposit to a stranger is a leap of faith

Car, real-estate and second-hand marketplaces share one weak spot: the up-front deposit.

1

A listing that looks too good

Car, flat or rental at a great price. The "seller" asks for a deposit to hold it.

2

Paid by bank transfer

Straight to a stranger's IBAN. No vault, no conditions, no undo.

3

Then the seller disappears

Listing deleted, phone offline. No history to check, nobody to chase.

POLISH CIVIL CODE

Art. 394

"zadatek" (earnest money)

Buyer backs out → **seller keeps the deposit.**

Seller backs out → **buyer gets double back.**

The law already says who pays.

But nobody sues a stranger over a few thousand złoty.

Who it's for — and what we redesigned

PRIMARY USER

People in Poland paying a zadatek to strangers from listing sites — and their sellers.

Not crypto users. The app says “account”, shows amounts in PLN and hides the chain.

SECONDARY USER

Marketplaces that embed the widget — our distribution channel.

1

Which relationship?

An up-front deposit paid to a stranger before the deal is completed.

2

Who was the intermediary?

Usually nobody — an IBAN transfer straight to the stranger. Or costly ones: a court to enforce art. 394, a notary escrow, a platform's paid payment protection.

3

What changes without it?

The rule runs instantly, with no court: whoever backs out loses; a seller who walks pays 2×. Money waits in the program, not with either side. No fee, and nobody can block the payout.

The law, as code

The deposit rule becomes a Solana program. Nobody in the middle holds the money.

- 1

Seller locks a stake **S** first

Equal to **D** — a fake seller must risk their own money.
- 2

Buyer locks the deposit **D**

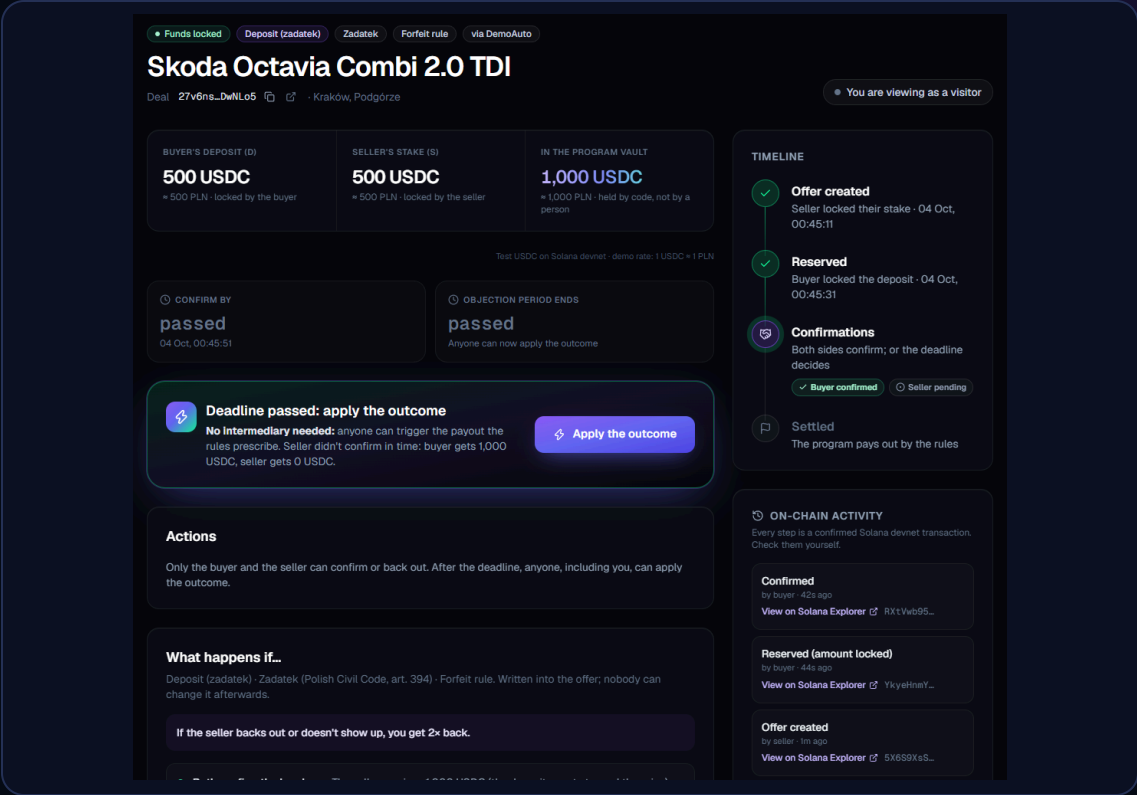
Into a program-owned vault — never the seller’s account.
- 3

Both confirm the handover

Two confirmations release $P = D + S$ to the seller.
- 4

Someone backs out or vanishes?

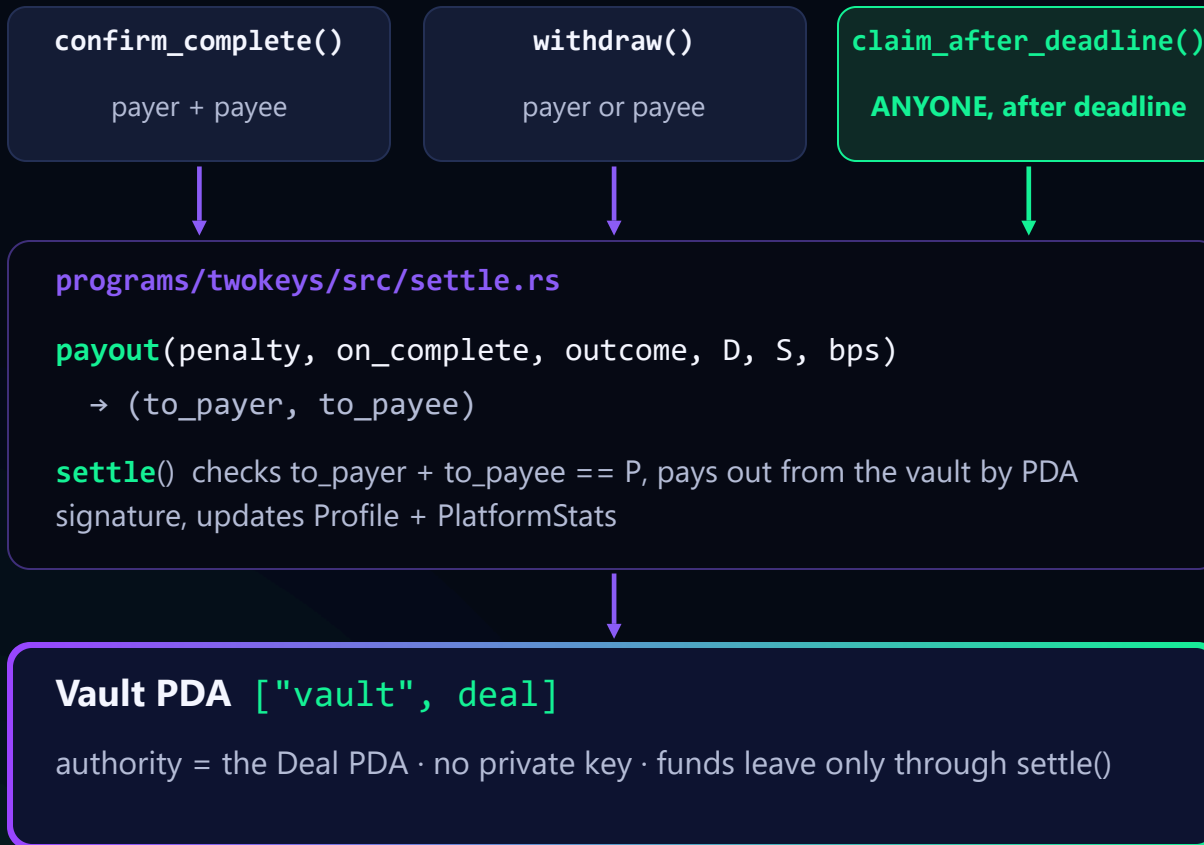
The rule pays out: if the seller walks, the buyer gets 2×.



/d/<deal> after the deadline: even a visitor can press “Apply the outcome”

Where the intermediary disappears

Every rule that moves money lives in the on-chain program. The app and widget cannot move funds.



THE MOMENT THE INTERMEDIARY IS NO LONGER NEEDED

Seller vanished? After the deadline, **anyone** can press "Apply the outcome" (claim_after_deadline). The program counts the missing party as a no-show and pays out by the rule. No court, no support ticket, no admin.

WHO CAN CALL WHAT

create_offer	payee
reserve	payer
confirm_complete · withdraw · open_dispute	payer or payee
resolve	arbiter, only if agreed
claim_after_deadline · expire_dispute	anyone
cancel_offer	payee; anyone after window

No admin instruction. Upgrade authority will be revoked after final deploy.

Same program, four templates, one payout table

Deposit · zadatek
payer / payee = buyer / seller
Forfeit · ToPayee · S=D

Rental
payer / payee = renter / owner
Forfeit · ToPayer · S=0

Freelance
payer / payee = client / freelancer
Forfeit · ToPayee · S=0

Purchase
payer / payee = buyer / seller
Forfeit · ToPayee · S=0

OUTCOME	WHEN	Forfeit	Refund
		to payer / to payee	to payer / to payee
Completed	Both confirmed (ToPayee · ToPayer)	0 / P · D / S	0 / P · D / S
PayerWithdrew / NoShow	Payer backed out, or only payee confirmed	0 / P	D / S
PayeeWithdrew / NoShow	Payee backed out, or only payer confirmed	P / 0 2× for zadatek	D / S
Expired	Deadline passed, nobody confirmed	D / S	D / S
Resolved	Optional arbiter splits by payer_bps	bps split	bps split
DisputeTimeout	Arbiter missed the deadline	D / S	D / S
Cancelled	No payer reserved the offer	– / S	– / S

D = payer amount · S = payee stake · P = D + S · payouts always sum to P · label Zadatek enforces Forfeit + ToPayee + S = D; Zaliczka enforces Refund

One component. Every marketplace, every sector.

1 Drop-in widget

```
<script src="https://<app>/widget.js"
  data-platform="<PLATFORM_PUBKEY>"
  data-listing-id="abc123"
  data-amount="2000"
  data-template="deposit"></script>
```

`data-template` = deposit | rental | freelance | purchase

2 TypeScript SDK

`createOffer()`

`reserve()`

`confirm()`

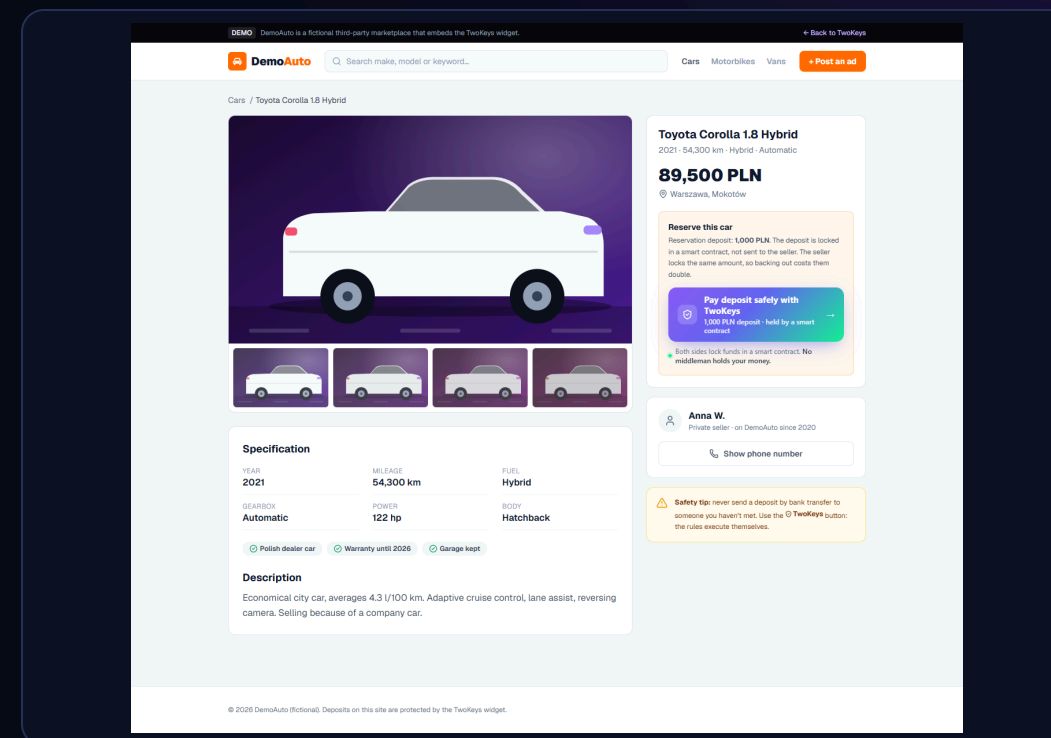
`withdraw()`

`claimAfterDeadline()`

`getProfile()`

3 Shareable deal link

`/d/<deal>` — send it on WhatsApp; both sides see the same live page.



DemoAuto · cars · deposit

DemoRent · cameras · rental

Fictional demo marketplaces — same widget, same program

Money on-chain. People off-chain.

GDPR-friendly by construction: reputation without identity.

ON-CHAIN

Solana · public · permanent

- Amounts, deadlines, rule parameters, status
- Pseudonymous wallet addresses
- Salted SHA-256 hashes of listing & evidence
- Behaviour counters per wallet (Profile)
- Anonymous platform aggregates (PlatformStats)

OFF-CHAIN

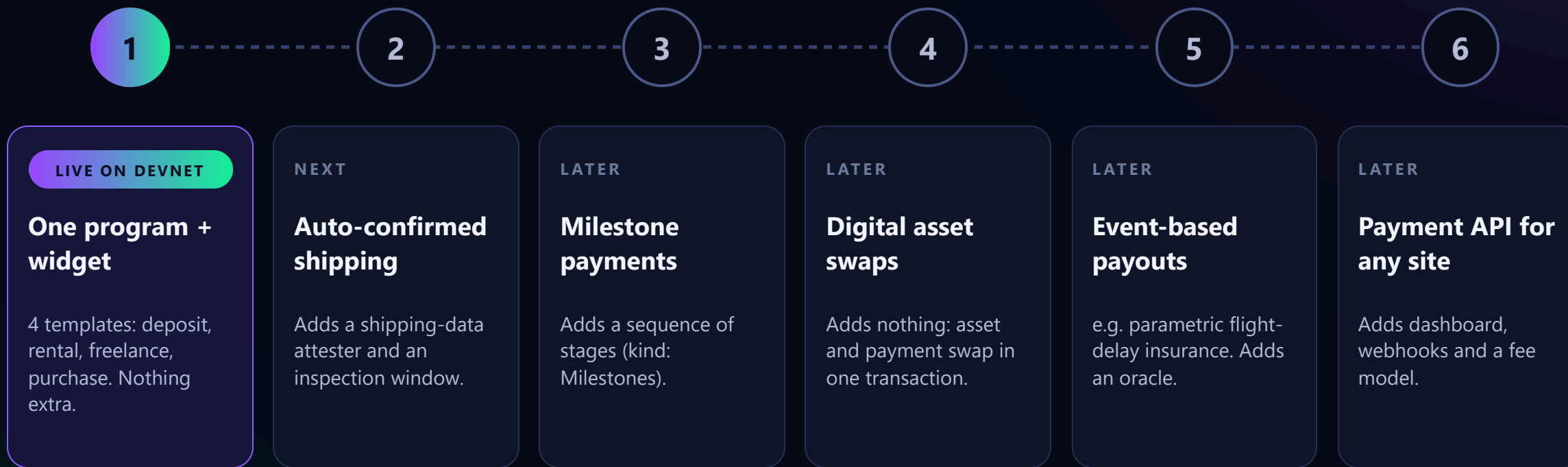
app server · private · deletable

- Names, emails, phone numbers
- Listing details, photos, evidence files
- Hash salts — delete one, the hash points to nothing
- Wallet ↔ person link
- **Off-chain data can't move money or outcomes**

Never on-chain: names · emails · phones · addresses · plates · listing text · free text

Deposits are step one

A trust layer any website can plug in — the same Deal account; a new kind only for new mechanics.



Deliberately out of scope: betting / prediction markets. **Vision only, not in the MVP:** encrypted identity escrow for large deals (an off-chain trust feature).

2K

One component — every marketplace, every sector, every country.

Solana

Anchor 0.32

SPL Token

Next.js

TypeScript SDK

REPO	github.com/89CF/twokeys
VIDEO	linked in the HackTribe submission (2:32)
LIVE DEMO	twokeys-psi.vercel.app · devnet, demo accounts built in
PROGRAM	AkQXPVXUYDqyNUVNAsYGYXy9sHQR636xcuJbAJkiJe5F
TEAM	Farazi

LIMITATIONS (HONEST)

- The handover is off-chain: confirmations come from the parties.
- An arbiter is optional — and trusted once both agree to one.
- Devnet and test USDC only; not audited, not production-ready.
- Legal effect of an on-chain zadatek needs legal review.